

Alexis Sáez

Postdoctoral Scholar

California Institute of Technology (Caltech)

Division of Geological and Planetary Sciences

1200 E California Boulevard, Pasadena, CA, USA 91125

Phone: (+1) 626 989 2182, E-mail: saez@caltech.edu

Research interests: My research focuses on understanding the physics of earthquakes and faults, with applications to the development of underground technologies with the potential to decarbonize our energy systems (e.g., deep geothermal energy and geological carbon storage) and to the mitigation of natural and human-induced hazards. I combine tools of theoretical and computational continuum mechanics with applied mathematics to develop physics-based models that are informed by and explain geological, geophysical, and laboratory observations. My current and past research topics include fluid-driven frictional ruptures, injection-induced seismicity, slow earthquakes and their connection to metamorphic fluids release, earthquake segmentation patterns of major plate boundaries, earthquake swarms dynamics, controls on earthquake nucleation timing, and long-term dynamics of fault networks, among others.

EDUCATION

EPFL - École Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

Ph.D. in Mechanics

2019-2023

- Advisor: Prof. Brice Lecampion.
- Research areas: Geomechanics, induced seismicity, earthquake and fracture mechanics, theoretical and computational mechanics.
- Awarded the Rocha Medal 2026 by the International Society for Rock Mechanics and Rock Engineering (ISRM) as the best PhD thesis in the field, worldwide, defended in 2023-2024.
- Awarded as the best PhD thesis in Mechanics in EPFL in 2024 (EDME award).

University of Chile

Santiago, Chile

M.Sc.E., B.Sc.E., and professional title in Civil Engineering

2005-2014

- Awarded as the Best Graduate Civil Engineer of the 2014 Class.

PROFESSIONAL EXPERIENCE

California Institute of Technology | Division of Geological and Planetary Sciences

CA, USA

Postdoctoral Scholar Research Associate in Geology

2024-Today

- Advisor: Prof. Jean-Philippe Avouac.
- Collaborators: Profs. Nadia Lapusta and Ares Rosakis.
- Research areas: Geophysics, earthquake source physics, slow and fast earthquakes, experimental fault mechanics, induced seismicity, geomechanics for geo-energy applications.
- Awarded the Swiss NSF Postdoc.Mobility Fellowship.
- Awarded the Caltech Geophysics Option Postdoctoral Fellowship.

Dalhousie University | Department of Civil and Resource Engineering

NS, Canada

Postdoctoral Fellow in Civil and Resource Engineering

2024

- Advisor: Prof. Dmitry Garagash.
- Research areas: Earthquake source physics, slow earthquakes, theoretical mechanics.

EPFL - École Polytechnique Fédérale de Lausanne | Institute of Civil Engineering

Lausanne, Switzerland

Doctoral Researcher in Mechanics

2019-2023

- Advisor: Prof. Brice Lecampion.
- Research areas: Geomechanics, induced seismicity, earthquake and fracture mechanics, theoretical and computational mechanics.
- Awarded the Swiss Government Excellence Scholarship for Foreign Students.

The University of Tokyo | Earthquake Research Institute

Tokyo, Japan

Visiting Doctoral Researcher in Geophysics

2022-2023

- Advisor: Prof. Aitaro Kato.
- Research areas: Geophysics, slow earthquakes.
- Awarded the EPFL Doc.Mobility Grant.

CODELCO - Corporación Nacional del Cobre de Chile

Santiago, Chile

Expert Consultant

2025-Today

- Consultancy on mining-induced seismicity and hydraulic fracturing operations.

Project Engineer

2013-2016

- Structural, geomechanical, and seismic engineering revisions for large mining projects and tailings dams.
- Management of engineering mining projects, techno-economic analysis, and strategic planning.

AWARDS

2025	Rocha Medal 2026: given by the International Society for Rock Mechanics and Rock Engineering (ISRM), to the best PhD thesis in Rock Mechanics and Rock Engineering worldwide, defended in 2023-2024.
2024	2024 EDME award: given by the Doctoral Program in Mechanics (which includes biomechanics, geomechanics, fluid mechanics, and solid mechanics) of the École Polytechnique Fédérale de Lausanne, to the best PhD thesis of the 2024 class.
2014	Best Graduate Engineer Award: given by the Chilean Society of Engineers (Colegio de Ingenieros de Chile) to the best graduate civil engineer of the 2014 class of the University of Chile.
2006, 2007, 2008, 2009, 2010	Outstanding Student Award: granted each year by the Faculty of Physical and Mathematical Sciences of the University of Chile, to undergraduate students belonging to the 6% of higher grades of the faculty.

GRANTS AND SCHOLARSHIPS

2026-2027	GMG Grant: granted by the Industrial Advisory Board of the Center for Geomechanics and Mitigation of Geohazards (GMG) at Caltech, to conduct research on the development of theoretical models of injection-induced fault slip. Amount: ~ 40,000 USD.
2025-2027	SNSF Postdoc Mobility Fellowship: granted by the Swiss National Science Foundation (SNSF) to conduct research at Caltech on slow earthquakes. Amount: ~ 155,000 USD.
2024-2025	Caltech Geophysics Option Postdoctoral Fellowship: granted by the Caltech Division of Geological and Planetary Sciences to work on the broad field of geophysics. Amount: ~ 70,000 USD.
2022-2023	EPFL Doc Mobility Grant: given by the École Polytechnique Fédérale de Lausanne (EPFL) to carry out a doctoral research stay of 6 months in the Earthquake Research Institute (ERI) of the University of Tokyo. Amount: ~ 10,000 USD.
2019-2023	Swiss Government Excellence Scholarship for Foreign Students: granted by the Swiss Confederation to conduct research at the École Polytechnique Fédérale de Lausanne (EPFL) as part of the PhD program in Mechanics. Amount: ~ 75,000 USD.
2022	EPFL-ETH Zürich Summer School Grant: given by EPFL and ETH Zürich to organize an international summer course for PhD students: “Mechanics of earthquakes and aseismic slip”, held in Zürich, Switzerland, July 2022. Amount: ~ 20,000 USD.
2011	Full Tuition Fee Waiver Scholarship: given by the Department of Civil Engineering, University of Chile, once per year to one outstanding student to study the master’s program in civil engineering. Amount: ~ 4,000 USD.

2005-2010 **Bicentenario Scholarship:** given by the Chilean government to students with financial difficulties and outstanding results in the admission exam to public universities. Amount: ~ 5,000 USD.

PUBLICATIONS

Journal Articles - published

1. **Sáez, A.**, Lecampion, B., Bhattacharya, P. and Viesca, R.C. (2022). Three-dimensional fluid-driven stable frictional ruptures. *Journal of the Mechanics and Physics of Solids*, 132, 103672. [link]
2. **Sáez, A.** and Lecampion, B. (2023). Post-injection aseismic slip as a mechanism for the delayed triggering of seismicity. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 479, 20220810. [link]
3. **Sáez, A.** and Lecampion, B. (2024). Fluid-driven slow slip and earthquake nucleation on a slip-weakening circular fault. *Journal of the Mechanics and Physics of Solids*, 183, 105506. [link]
4. **Sáez, A.**, Passelègue, F. and Lecampion, B. (2025). Maximum size and magnitude of injection-induced slow slip events. *Science Advances*, 11, eadq0662. [link]

Journal Articles - in review

5. Lecampion, B., Brisson, S., Sarma, A., Gupta, A., **Sáez, A.**, Fakhretdinova, R. (2026). Three dimensional simulation of fluid-driven frictional and tensile ruptures on existing discontinuities. *ArXiv*. [link]

Journal Articles - in preparation

6. **Sáez, A.**, Kato, A., and Garagash, D. (2026). Slow slip events driven by episodic fluid release from slab dehydration. *In prep. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
7. **Sáez, A.**, and Avouac, J.-P. (2026a). Crack models for earthquake barriers. *In prep. [presented as part of the invited talk: Towards a mechanistic understanding of fault rupture segmentation at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
8. **Sáez, A.**, and Avouac, J.-P. (2026b). Fault segmentation can be an illusion. *In prep. [presented as part of the invited talk: Towards a mechanistic understanding of fault rupture segmentation at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
9. **Sáez, A.**, Lapusta, N., and Avouac, J.-P. (2026). Leak-off effects on fault-zone fluid transport and injection-induced aseismic slip. *In prep. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
10. **Sáez, A.**, Sarma, A., Avouac, J.-P., and Lecampion, B. (2026). Spatiotemporal patterns of earthquake swarms driven by fluid injections and induced aseismic slip. *In prep. [preliminary results presented at EGU General Assembly, Vienna, Austria, May 23-27, 2022]* [link]
11. Rodríguez-Padilla, A., **Sáez, A.**, and Avouac, J.-P. (2026). Earthquake sequence synchronization and alternation driven by quasistatic elastic interactions across faults. *In prep. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
12. Shi, Q., **Sáez, A.**, Li, L., Im, K., Bourne, S., van Elk, J., and Avouac, J.-P. (2026). Conditions for run-away ruptures during reservoir operations, implication for the magnitude-frequency distribution of induced earthquakes and application to Groningen. *In prep. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]
13. Hernández-Montenegro, J., **Sáez, A.**, Dong, J., and Asimow, P. (2026). Intrusive magmatism as a potential source of mid-crustal water on Mars. *In prep. [presented at AGU Fall Meeting, New Orleans, LA, US, December 15-19, 2025]* [link]

14. Gupta, A., **Sáez, A.**, and Lecampion, B. (2026). Shearing and opening of a fault due to fluid injection. *In prep. [presented at the 26th International Congress of Theoretical and Applied Mechanics, Daegu, Korea, August 25-30, 2024]*
15. Fakhretdinova, R., **Sáez, A.**, Alcolea, A., and Lecampion, B. (2026). Revisiting the Basel-1 hydraulic stimulation with a 3D coupled hydro-mechanical model. *In prep. [preliminary results presented at 50th Stanford Geothermal Workshop, CA, USA, February 10-12, 2025] [link]*
16. Egorov, S., **Sáez, A.**, Lecampion, B., Lebihain, M. (2026). A perturbation approach for fluid-driven frictional shear crack propagation. *In prep.*
17. Marty, S., Avouac, J.-P., Curry, W., **Sáez, A.**, da Silva, J. (2026). Earthquakes induced by hydraulic fracturing, a semi-analytical model with application to the Permian Basin. *In prep.*

Proceedings/Conference Articles

1. Lecampion, B., **Sáez, A.** and Gupta, A. (2023). Shearing and opening of a pre-existing discontinuity in response to fluid injection. *57th U.S. Rock Mechanics/Geomechanics Symposium*, Atlanta, Georgia, USA, June 25-28. [link]
2. Fakhretdinova, R., **Sáez, A.**, Alcolea, A., and Lecampion, B. (2025). Revisiting the Basel-1 hydraulic stimulation with a 3D coupled hydro-mechanical model. *50th Stanford Geothermal Workshop, CA, USA, February 10-12. [link]*

PhD Thesis

1. **Sáez, A.** (2023). Three-dimensional fluid-driven frictional ruptures: theory and applications. PhD thesis, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland. [link]

TEACHING AND MENTORING EXPERIENCE

California Institute of Technology | Division of Geological and Planetary Sciences

• *PhD co-supervision*: “Physical controls on earthquake maximum magnitude at the Groningen gas field”. Student: Qian Shi.

2025-2026

EPFL - École Polytechnique Fédérale de Lausanne | Institute of Civil Engineering

- *Teaching Assistant for Computational Geomechanics (CIVIL-423)*: Master's course on poroelasticity theory and its numerical solution through the finite element method. 2019-2021
- *Teaching Assistant for Geotechnical Engineering (CIVIL-306)*: Bachelor's course on fundamentals of soil mechanics and engineering design of soil structures. 2020-2021
- *PhD co-supervision*: "Effects of permeability variations and dilatancy on injection-induced fault slip". Student: Regina Fakhretdinova. 2021-2025
- *Master's final project supervision*: "Revisiting the Pohang induced earthquake". Student: Tristan Liardon (currently, PhD student at EPFL). 2021
- *Bachelor and Master's semester projects supervision*: 2 bachelors, 2 masters. 2020-2022

University of Chile | Departments of Physics, Mathematics, and Civil Engineering

- **Physics Department**: *Teaching Assistant for Newtonian Mechanics (bachelor level), and Statistical Physics (bachelor level)*. 2007-2009
- **Mathematics Department**: *Teaching Assistant for Differential Calculus (bachelor level), Ordinary Differential Equations (bachelor level)*. 2007-2009
- **Civil Engineering Department**: *Teaching Assistant for Solid Mechanics I (bachelor level), Solid Mechanics II (master level), Structural Analysis (bachelor level), Structural Dynamics (master level), Non-linear Analysis of Structures (master level)*. 2009-2014

SERVICE

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| 2022 - now | Peer reviewer for Journal of Geophysical Research: Solid Earth, Earth and Planetary Science Letters, Journal of Fluid Mechanics, Philosophical Transactions of the Royal Society A, Geophysical Journal International, Rock Mechanics and Rock Engineering, Communications Earth and Environment. |
| 2025 - now | Member of the Brown Bag Seminar organizing committee at the Seismological Laboratory of Caltech. |
| 2022 | Creator and principal organizer of the international summer course for PhD students: "Mechanics of earthquakes and aseismic slip", Zürich, Switzerland, 18-21 July 2022. |
| 2016-2017 | Member of the technical committee CL034 of the National Standardization Institute of Chile for the development of the national code NCh3411 for the seismic design of buildings with passive energy dissipation systems. |
| 2012 | Member of conference organizing committee: Behavior of Steel Structures in Seismic Areas (STESSA) 7th International Conference, Santiago, Chile. |
| 2010 | Member of conference organizing committee: Proceedings of the 10th Chilean Conference on Seismology and Earthquake Engineering, Santiago, Chile. |

TECHNOLOGY TRANSFER

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| 2019-Today | Co-development of a C++ computational code for the solution of vectorial Boundary Integral Equations accelerated with hierarchical matrices. [link] |
| 2014 | Development of a proposal for the national code for the seismic design of buildings with passive energy dissipation systems. This proposal was the main output of my master's thesis and provided the basis for the now official national code NCh3411 of the National Standardization Institute of Chile. [link] |

INVITED TALKS

- 04/2026 SZ4D Science Meeting 2026. “Slow slip events driven by episodic fluid release from slab dehydration”, Long Beach, LA, US.
- 04/2026 SSA Annual Meeting 2026, Session: Earthquake swarms across the earthquake cycle: precursors, transients and after effects. “Spatio-temporal migration patterns of micro-seismicity driven by fluid injections and induced aseismic slip”, Pasadena, LA, US.
- 03/2026 Department of Earth Sciences, University of Southern California - Lithosphere Dynamics Seminar Series. “Slow slip events driven by episodic fluid release from slab dehydration”, Los Angeles, CA, USA.
- 12/2025 AGU Fall Meeting 2025, Session S23B - Kinematic, Dynamic, and Analytical Earthquake Source Modeling. “Towards a mechanistic understanding of fault rupture segmentation”, New Orleans, LA, US.
- 09/2025 Chilean Society of Rock Mechanics. “Fluid-driven shear ruptures: theory and applications to fluid-induced seismicity”, Santiago, Chile.
- 09/2025 American Rock Mechanics Association (ARMA) - Future Leaders Webinar Series. “Fluid-driven shear ruptures: theory and applications to fluid-induced seismicity”, online.
- 05/2025 École Polytechnique Fédérale de Lausanne - On the occasion of the Mechanics Day 2025 to receive the EDME award for the best PhD thesis. “Fluid-driven frictional ruptures: theory and applications to fluid-induced seismicity”, Lausanne, Switzerland.
- 05/2025 Scripps Institution of Oceanography, UC San Diego - Geophysics Seminar Series. “Fluid-driven shear ruptures: theory and applications to fluid-induced seismicity”, Online.
- 04/2025 Seismological Laboratory, Caltech - Seismolab Seminar Series. “Fluid-driven shear ruptures: theoretical models and their applications to fluid-induced seismicity”, Pasadena, CA, USA.
- 03/2025 Earth and Planetary Sciences Department, UC Santa Cruz - IGPP Seminar Series. “Fluid-driven shear ruptures: theoretical models and their applications to fluid-induced seismicity”, Santa Cruz, CA, USA.
- 11/2024 Department of Earth Sciences, University of Southern California - Lithosphere Dynamics Seminar Series. “Fluid-driven frictional ruptures: theory and applications to injection-induced seismicity”, Los Angeles, CA, USA.
- 03/2024 GeoREST-Workshop on induced seismicity. “Reactivation of fault slip during and after fluid injections: theoretical considerations, micro-seismicity patterns, and aseismic moment release.” Palma de Mallorca, Spain.
- 02/2023 Seismology and Rock Mechanics groups, National Research Institute for Earth Science and Disaster Prevention (NIED), Japan. “Fluid-driven stable fault slip: nucleation, propagation and arrest”, Tsukuba, Japan.
- 02/2023 Seismology and Physics of the Earth’s interior group, Department of Geophysics, Kyoto University. “Fluid-driven stable fault slip: nucleation, propagation and arrest”, Kyoto, Japan.
- 08/2022 Seismological Laboratory, Caltech - Brown Bag Seminar Series. “Fluid-driven stable fault slip: nucleation, propagation and arrest”, Pasadena, CA, USA.
- 08/2022 Earth Resources Laboratory, MIT. “Fluid-driven stable fault slip in 3D: nucleation, propagation and arrest”, Cambridge, MA, USA.

- 08/2022 Schlumberger-Doll Research. “Fluid-driven stable fault slip: nucleation, propagation and arrest”, Cambridge, MA, USA.
- 05/2022 Geoazur Laboratory, Université de Côte d’Azur. “Post-injection pulse-like aseismic ruptures”, Sophia Antipolis, France.
- 06/2021 Applied and Earthquake Seismology Research Group, Department of Earth Sciences, Freie Universität Berlin. “Three-dimensional aseismic ruptures driven by fluid injection”, Berlin, Germany. Online.
- 05/2021 Seismology Group, Department of Geophysics, University of Chile,. “Three-dimensional aseismic ruptures driven by fluid pressure diffusion”, Santiago, Chile. Online.
- 04/2021 GFZ German Research Centre for Geosciences - Section 4.1 Lithosphere Dynamics. “Three-dimensional aseismic ruptures driven by fluid injection”, Potsdam, Germany. Online.

ORAL PRESENTATIONS

- 12/2025 AGU Fall Meeting 2025, Session S11A - Fluids, Faulting and Earthquakes: The Role of Fluids in Seismogenesis. “Fluid flow along fault zones including leak-off into host rocks: implications for injection-induced fault slip and pore pressure distribution at depth along plate-boundary faults”, New Orleans, LA, US.
- 10/2025 VII Colloquium on Geophysical Signatures of Earthquakes and Volcanoes. “Towards a mechanistic understanding of fault rupture segmentation”, Valparaíso, Chile.
- 01/2025 Slow-to-fast Earthquake Workshop in Chile. “Slow-slip events as a fluid-driven frictional rupture process”, Santiago, Chile.
- 10/2024 Cargèse International Workshop on Earthquakes, 4th Edition. “Maximum size and magnitude of injection-induced slow slip events”, Cargèse, Corsica, France.
- 05/2024 Japan Geoscience Union (JpGU) - Meeting 2024. “Slow earthquakes and pore pressure oscillations”, Makuhari Messe, Chiba Prefecture, Japan. Online.
- 12/2023 American Geophysical Union (AGU) - Fall Meeting 2023. “Spatio-temporal patterns of seismic swarms driven by coupled pore pressure diffusion and aseismic slip events”, San Francisco, CA, USA. Online.
- 05/2023 European Geosciences Union (EGU) - General Assembly 2023. “Extraction of fluids to mitigate the seismic risk associated with post-injection aseismic slip”, Vienna, Austria.
- 12/2022 American Geophysical Union (AGU) - Fall Meeting 2022. “Pulse-like aseismic ruptures upon the cessation of fluid injections: regimes of propagation and arrest in 3D”, Chicago, IL, USA. Online.
- 07/2022 11th European Solid Mechanics Conference (ESMC). “Three-dimensional fluid-driven stable frictional ruptures”, Galway, Ireland.
- 05/2022 European Geosciences Union (EGU) - General Assembly 2022. “Spatio-temporal patterns of fluid-driven aseismic slip transients: implications for seismic swarms”, Vienna, Austria.
- 05/2022 European Geosciences Union (EGU) - General Assembly 2022. “2D and 3D numerical modeling of fluid-driven frictional crack growth for geothermal hydraulic stimulation”, Vienna, Austria.
- 05/2021 European Geosciences Union (EGU) - General Assembly 2021. “Three-dimensional aseismic ruptures driven by fluid injection”, Vienna, Austria. Online.

PROGRAMMING LANGUAGES

Proficient in C++, Python, Wolfram Mathematica, and Matlab.

LANGUAGES

Fluent in English, Spanish (native), and Portuguese.

PERSONAL DEVELOPMENT

2017-2018 Trip around the world (Australia, China, Brazil, Myanmar, among many other countries).

REFERENCES

Brice Lecampion

Associate Professor, Head of Geo-Energy Laboratory,
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland.
E-mail: brice.lecampion@epfl.ch

Jean-Philippe Avouac

Earle C. Anthony Professor of Geology and Professor of Mechanical and Civil Engineering,
California Institute of Technology.
E-mail: avouac@caltech.edu

Eric M. Dunham

Professor of Geophysics,
Stanford University.
Email: edunham@stanford.edu

Ares Rosakis

Theodore von Kármán Professor of Aeronautics and Professor of Mechanical Engineering,
California Institute of Technology.
Email: arosakis@caltech.edu

Nadia Lapusta

Lawrence A. Hanson, Jr., Professor of Mechanical Engineering and Geophysics,
California Institute of Technology.
Email: lapusta@caltech.edu

Aitaro Kato

Professor, Earthquake Research Institute, The University of Tokyo.
Email: akato@eri.u-tokyo.ac.jp

Dmitry Garagash

Professor of Civil and Resource Engineering,
Dalhousie University.
Email: garagash@dal.ca